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Identification of NLRP3 Inflammasome-related biomarkers in PPMI CSF Samples
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Summary (or Abstract)

Introduction: The NOD-like receptor protein 3 (NLRP3) inflammasome plays an important role in Parkinson's disease (PD) pathogenesis (Heneka et al., 2013, Haque et al., 2020). However, the biological levels of NLRP3-related biomarkers in PD CSF is largely unknown. In this study, we employed a high sensitivity SIMOA-based IL-1b assay to determine baseline and follow-up levels of IL-1b in controls and individuals with PD.

Results: IL-1b could be detected with high sensitivity in the majority of samples with only n=3 samples being BLQ. No significant differences were observed when comparing IL-1b levels between groups. No longitudinal changes in IL-1b levels were detected.

Conclusions: High sensitivity SIMOA-based assays support the assessment of otherwise notoriously difficult to measure analytes. IL-1b levels do not seem to be different between PD and controls.

Limitations: The study made use of remaining sample volumes from a previous PPMI project. The samples were outside of the determined long-term stability data for this assay which may have impacted data quality.

Methods

Simoa (Single Molecular Array) technology developed by Quanterix allows detection of thousands of single protein molecules simultaneously. Utilizing the same reagents as a conventional ELISA, this method has been used to measure proteins in a variety of different matrices (serum/plasma, cerebrospinal fluid, urine, cell extracts etc.) at femtomolar (fg/ml) concentrations, offering a roughly 1000-fold improvement in sensitivity. This approach makes use of arrays of femtoliter-sized reaction chambers, termed single-molecule arrays (SimoaTM) that can isolate and detect single enzyme molecules. Because the array volumes are approximately 2 billion times smaller than a conventional ELISA, a rapid buildup of a fluorescent product is generated if a labeled protein is present. With diffusion defeated, this high local concentration of product can be readily observed. Only one single molecule is needed to reach the detection limit.

In the first step of this single-molecule immunoassay, antibody capture agents are attached to the surface of paramagnetic beads (2.7 mm diameter) that will be used to concentrate a dilute solution of molecules. The beads typically contain approximately 250,000 attachment sites so one can think of each bead as having a "lawn" of capture molecules. The beads are added to the sample solution such that there are many more beads than target molecules. Typically, 500,000 beads will be added to a 100 µL sample. There are two advantages to adding so many beads: first, at a roughly 10:1 bead to molecule ratio, the percentage of beads that contain a labeled immunocomplex follows a Poisson distribution. At low concentrations of protein, the Poisson distribution indicates that each bead will capture either a single immunocomplex or none. For example, if 1 fM of a protein in 0.1 ml (60,000 molecules) is captured and labeled on 500,000 beads, then 12% of the beads will carry one protein molecule and 88% will not carry any protein molecules. Second, with so many beads in solution, the bead-to-bead distance is small, such that every molecule encounters a bead in less than a minute. Diffusion of the target analyte molecules, even large proteins, occurs on a time scale such that all the molecules should in theory quickly have multiple collisions with multiple beads. In this manner, the slow binding to a fixed capture surface is avoided and the efficiency of binding increases dramatically. Beads are then washed to remove nonspecifically bound proteins, incubated with biotinylated detection antibody and then with β-galactosidase labeled streptavidin. In this manner, each bead that has captured a single protein molecule is labeled with an enzyme. Beads that do not capture a molecule remain label free.

Rather than ensemble readout, beads are loaded into arrays of 216,000 femtoliter-sized wells that have been sized to hold no more than one bead per well (4.25 mm width, 3.25 mm depth). Beads are added in the presence of substrate, and wells are subsequently sealed with oil and imaged. Simoa permits the detection of very low concentrations of enzyme labels by confining the fluorophores generated by individual enzymes to extremely small volumes (~40 fL), ensuring a high local concentration of fluorescent product molecules. If a target analyte has been captured (immunocomplex formed), then the substrate will be converted to a fluorescent product by captured enzyme label. The ratio of the number of wells containing a bead with an enzyme label to the total number of wells containing a bead corresponds to the analyte concentration in the sample. By acquiring two fluorescence images of the array, it is possible to demonstrate an increase in signal thereby confirming the presence of a true immunocomplex, and those beads associated with a single enzyme molecule ("on" well) can be distinguished from those not associated with an enzyme ("off" well). The protein concentration in the test sample is determined by counting the number of wells containing both a bead and fluorescent product relative to the total number of wells containing beads. As Simoa enables concentration to be determined digitally rather than by using the total analog signal, this approach to detecting single immunocomplexes has been termed digital

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ELISA. The ability of digital ELISA to measure much lower concentrations of proteins than conventional ELISA derives from two effects: (i) the high sensitivity of Simoa to enzyme labels and (ii) the low background signals that can be achieved by digitizing the detection of proteins. For antibodies of given affinity, the sensitivity of the immunoassay will be determined by the assay background. The high label sensitivity and decreased label concentration helps reduce nonspecific binding to the capture surface, resulting in much lower background signals.

The assay was developed for F Hoffmann La Roche on the SIMOA platform and run at Myriad. The LDD and LLOQ were 0.026 pg/ml and 0.018 pg/ml respectively.
Only n=3 individuals were BLQ and excluded from the analysis. Disparities in sample numbers as compared to the number of samples provided from previous grant ID 11872 are explained by the fact that not all samples had sufficient volume to run the test.

Statistical analysis:

All statistical analysis were performed with R. Measurements are log2 transformed.

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